

- Description

This module manages a liquid level monitoring system, ensuring that:

1. **The liquid level in a storage tank is continuously measured and checked against safety limits.**
2. **If the liquid level exceeds the maximum threshold, an overflow warning is triggered.**
3. **If the level is too low, a refill recommendation is generated to maintain operational stability.**
4. **A log of all level readings, warnings, and refill actions is maintained for safety compliance.**

This system helps prevent liquid overflow incidents and ensures a stable supply in industrial applications.

- Specification

module LiquidLevelMonitoringEx

var currentLevel, maxLevel, minLevel: real

var levelLog: sequence of string

process updateLevel(changeAmount: real)

functional scenario form:

currentLevel >= minLevel && levelLog = levelLog $\oplus$ ("Level Updated: " + currentLevel + " L"), Liquid level updated successfully.

||

currentLevel > maxLevel && levelLog = levelLog $\oplus$ ("Overflow Warning: " + currentLevel + " L"), Overflow warning triggered.

||

currentLevel < minLevel && levelLog = levelLog $\oplus$ ("Low Level Alert: " + currentLevel + " L"), Low level alert triggered.

end process

end module

- Java Code

- Coverage report

1. Test case number: 100
2. V-Method testing branch coverage: 83.3%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ c119_LiquidLevelSystemEx.java	81.0 %	47	11	58
> LiquidPolicy	50.0 %	1	1	2
> LevelStats	75.0 %	3	1	4
▼ LiquidTankEx	83.3 %	15	3	18
escapeCsv(String)	50.0 %	3	3	6
exportCsv(String)	100.0 %	4	0	4
printLevelLog()	100.0 %	2	0	2
updateLevel(double)	100.0 %	6	0	6
LiquidTankEx(LiquidPolicy, double)		0	0	0
computeStats()		0	0	0
> LevelAnomalyDetector	86.7 %	26	4	30
> TestSuites119	100.0 %	2	0	2
> c119_LiquidLevelSystemEx		0	0	0
> LevelRecord		0	0	0

3. Vibration testing instruction coverage: 96.5%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ c119_LiquidLevelSystemEx.java	44.7 %	812	1,005	1,817
> TestSuites119	7.2 %	71	920	991
> LevelAnomalyDetector	91.9 %	193	17	210
> LevelStats	92.6 %	162	13	175
> c119_LiquidLevelSystemEx	92.7 %	115	9	124
> LiquidPolicy	94.4 %	34	2	36
▼ LiquidTankEx	96.5 %	219	8	227
escapeCsv(String)	60.0 %	12	8	20
LiquidTankEx(LiquidPolicy, double)	100.0 %	14	0	14
computeStats()	100.0 %	4	0	4
exportCsv(String)	100.0 %	77	0	77
printLevelLog()	100.0 %	20	0	20
updateLevel(double)	100.0 %	92	0	92
> LevelRecord	100.0 %	18	0	18

4. Fuzz testing branch coverage: 83.3%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ c119_LiquidLevelSystemEx.java	81.0 %	47	11	58
> LiquidPolicy	50.0 %	1	1	2
> LevelStats	75.0 %	3	1	4
▼ LiquidTankEx	83.3 %	15	3	18
escapeCsv(String)	50.0 %	3	3	6
exportCsv(String)	100.0 %	4	0	4
printLevelLog()	100.0 %	2	0	2
updateLevel(double)	100.0 %	6	0	6
LiquidTankEx(LiquidPolicy, double)		0	0	0
computeStats()		0	0	0
> LevelAnomalyDetector	86.7 %	26	4	30
> TestSuites119	100.0 %	2	0	2
> c119_LiquidLevelSystemEx		0	0	0
▼ LevelRecord		0	0	0
LevelRecord(double, double, double)		0	0	0

5. Fuzz testing instruction coverage: 96.5%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ c119_LiquidLevelSystemEx.java	47.2 %	1,168	1,309	2,477
> TestSuites119	25.9 %	427	1,224	1,651
> LevelAnomalyDetector	91.9 %	193	17	210
> LevelStats	92.6 %	162	13	175
> c119_LiquidLevelSystemEx	92.7 %	115	9	124
> LiquidPolicy	94.4 %	34	2	36
▼ LiquidTankEx	96.5 %	219	8	227
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printLevelLog()	100.0 %	20	0	20
updateLevel(double)	100.0 %	92	0	92
▼ LevelRecord	100.0 %	18	0	18
LevelRecord(double, double, double)	100.0 %	18	0	18

6. Z3 branch coverage: 83.3%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ c119_LiquidLevelSystemEx.java	81.0 %	47	11	58
> LiquidPolicy	50.0 %	1	1	2
> LevelStats	75.0 %	3	1	4
▼ LiquidTankEx	83.3 %	15	3	18
escapeCsv(String)	50.0 %	3	3	6
exportCsv(String)	100.0 %	4	0	4
printLevelLog()	100.0 %	2	0	2
updateLevel(double)	100.0 %	6	0	6
LiquidTankEx(LiquidPolicy, double)		0	0	0
computeStats()		0	0	0
> LevelAnomalyDetector	86.7 %	26	4	30
> TestSuites119	100.0 %	2	0	2
> c119_LiquidLevelSystemEx		0	0	0
▼ LevelRecord		0	0	0
LevelRecord(double, double, double)		0	0	0

7. Z3 instruction coverage: 96.5%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ c119_LiquidLevelSystemEx.java	47.2 %	1,168	1,309	2,477
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exportCsv(String)	100.0 %	77	0	77
printLevelLog()	100.0 %	20	0	20
updateLevel(double)	100.0 %	92	0	92
▼ LevelRecord	100.0 %	18	0	18
LevelRecord(double, double, double)	100.0 %	18	0	18

8. Unimproved V-method branch coverage: 83.3%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ c119_LiquidLevelSystemEx.java	81.0 %	47	11	58
> LiquidPolicy	50.0 %	1	1	2
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> LevelAnomalyDetector	86.7 %	26	4	30
> TestSuites119	100.0 %	2	0	2
> c119_LiquidLevelSystemEx		0	0	0
> LevelRecord		0	0	0

9. Unimproved V-method instruction coverage: 96.5%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
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